Optimization III Answers

1. The minimum ladder length is about $\sqrt{224.77} \approx 15$ ft.
2. $h = 2R/\sqrt{3}, r = \sqrt{2/3}R$
3. (a) The objective function is $A(x) = \frac{x^2}{16} + \frac{\sqrt{3}}{36}(2-x)^2$, where $0 \leq x \leq 2$. This has a single critical point at $x = \frac{8\sqrt{3}}{9+4\sqrt{3}}$, which gives the minimum area. The maximum area will arise if we take the whole piece of wire and bend it into a square.
4. The objective function is $A(\theta) = 400(\sin(\theta) + \sin(\theta)\cos(\theta))$, where $0 \leq \theta \leq \frac{\pi}{2}$. The maximum volume is given by taking $\theta = \frac{\pi}{3}$.